ERIC HOU

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EDUCATION

PURDUE UNIVERSITY Expected Dec 2027
B.S. Electrical Engineering | Global Engineering Studies Minor GPA: 3.85 / 4.00

EXPERIENCE

Full Stack Robotics Engineering Intern May 2026 - Aug 2026
CircuitHub
- Designed and validated electromechanical test systems for precision pick-and-place hardware, integrating sensors, actuators, pneumatic controls, and PLC-based motion to diagnose failures, recovering $50K+ in hardware costs
- Architected control electronics for next-generation PnP voice-coil actuators, evaluating current regulation, position feedback, actuator response, and closed-loop behavior
- Automated hardware characterization with Python-to-TwinCAT ADS, analyzing current, force, position, and fault data across 30,000+ motion cycles to diagnose intermittent sensor and actuator failures

Avionics Engineer Aug 2025 - Present
Active Controls | Purdue Space Program (PSP)
- Designed STM32H7 flight-control PCB interfacing with the sensor suite for TOAD, a self-stabilizing VTVL rocket vehicle
- Developed buck-converter power distribution and monitoring boards for ASTRA and TOAD with CAN FD interfacing
- Integrated thermocouples, PTs, force cells, valves, and instrumentation for thruster test stands at Zucrow Propulsion Labs

Hardware Lead Jul 2021 - Jun 2025
FIRST Robotics Team 13356
- Led end-to-end robotics hardware development, including CAD, manufacturing, and validation across seasonal competitions
- Guided team to #3 in California, Top 30 worldwide; Qualified for MTI (top 0.5% of 7,000+ teams) three years running

RESEARCH

Analog / Mixed-Signal Researcher Jan 2026 - Present
System-on-Chip Extension Technologies (SoCET) - Vertically Integrated Projects at Purdue
- Implemented a closed-loop gm-boosted amplifier in Cadence Virtuoso for large capacitive loads, using transistor-level AC and transient simulations to characterize GBW, phase margin, slew rate to push the load-dependent output pole to higher frequency
- Developed SystemVerilog matrix coprocessor supporting up to 9×9 matrix addition, multiplication, transpose, and scalar operations, integrating a 33-state controller and multi-cycle arithmetic FSMs

Undergraduate Robotics Research Assistant Dec 2025 - May 2026
TRACE Lab (Terrain Robotics Advanced Control and Experimentation)
- Developed STM32 firmware with ADC and DMA drivers to interface soft sensors with ROS for humanoid haptic feedback

First Author - VertiPaw: Development of a Vertical Climbing Robot for Infrastructure Monitoring Sep 2023 - Dec 2024
IEEE International Conference on Robotic Computing (IRC)
- Designed and Authored VertiPaw, presented at IEEE IRC 2024, as the first quadrupedal robot designed specifically for vacuum-assisted vertical climbing across varied surfaces with heavy loads, achieving 9 lb payload capacity
- Integrated vacuum adhesion and servo-control electronics with a lightweight carbon-fiber, polycarbonate, and 3D-printed chassis
- Implemented LiDAR-based 3D localization and inverse-kinematics trajectory planning for stable multi-surface traversal

PROJECTS

3D Volumetric Display Jul 2026 - Sep 2026
- Engineered a 3D persistence-of-vision volumetric display integrating STM32WB55 control, BLDC motor drive, Bluetooth Low Energy (BLE) with a custom 2.4 GHz PCB antenna and RF matching network, and power charging / distribution subsystem
- Architected 1:12 multiplexed scanning for a 12×24 rotating RGB array, enabling efficient high-speed control of 288 RGB pixels

5-Stage Electromagnetic Coil Launcher Jan 2025 - May 2025
- Designed a multistage electromagnetic launcher with MOSFET-switched discharge PCBs handling measured 43 A pulses
- Implemented digitally controlled boost-converter charging to vary supercapacitor-bank voltage and control launch distance

Autodesk Fusion 360 Wire Generator Plugin Oct 2023 - Feb 2024
- Built a Fusion 360 plugin for parametric wire routing, generating and sweeping configurable 3D splines around bodies using geometric constraints and Rodrigues vector rotation
- Delivered a production-ready extension adopted by 5,000+ users worldwide and rated 5 stars on the Autodesk App Store

SKILLS

Hardware / EDA: Altium Designer, KiCad, Cadence Virtuoso, Fusion 360 CAD/CAM/FEA, SolidWorks, CNC, 3D Printing
Embedded / Digital: STM32, SystemVerilog, Verilog HDL, FPGA, SPI/UART/CAN, QuestaSim
Software: C, Python, MATLAB, TwinCAT XAE, Git